

PARAMETRIC PAVILION

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Assignment 02

Arch 565 - Advanced Computer Applications II

Spring 2026



EXISTING SITE

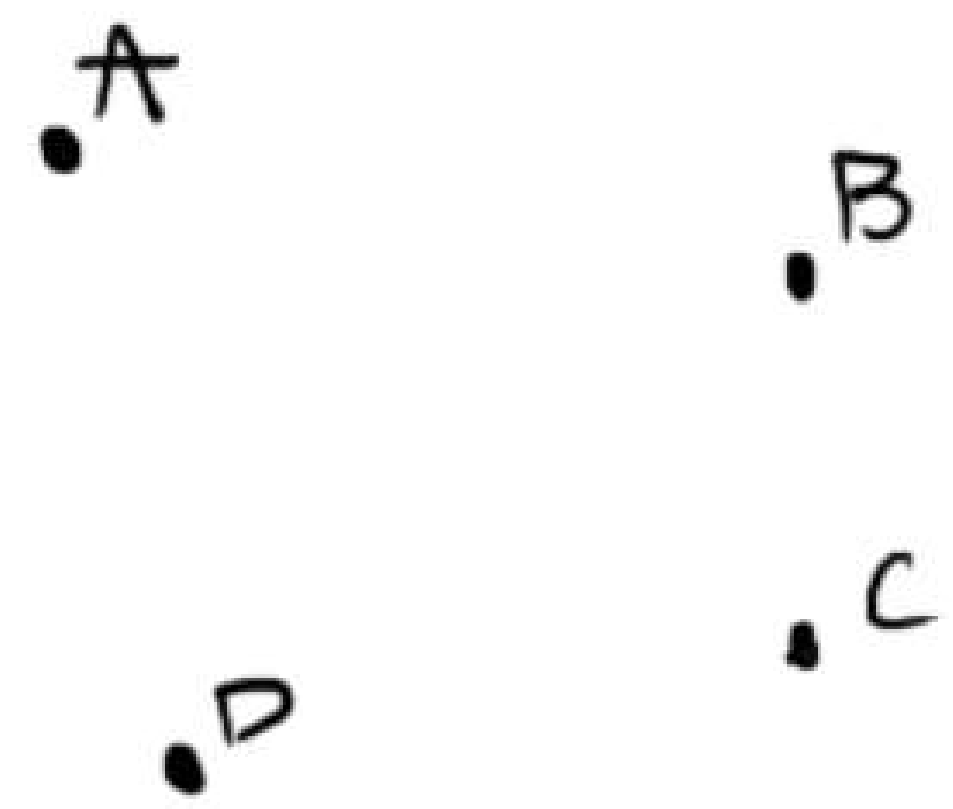
Rivers Edge Trail - Caboose Trailhead
Great Falls, MT



The River's Edge Trail is a major recreational corridor along the river in Great Falls, with the Caboose Trailhead as a key landmark. Currently dilapidated and underused, the site calls for a new architectural intervention. Defined by strong, persistent winds, the design proposes a pavilion that both celebrates and makes the wind visible. Positioned at the intersection of Great Falls, wind, and water, the project engages these forces while introducing new amenities and revitalizing a neglected public node.

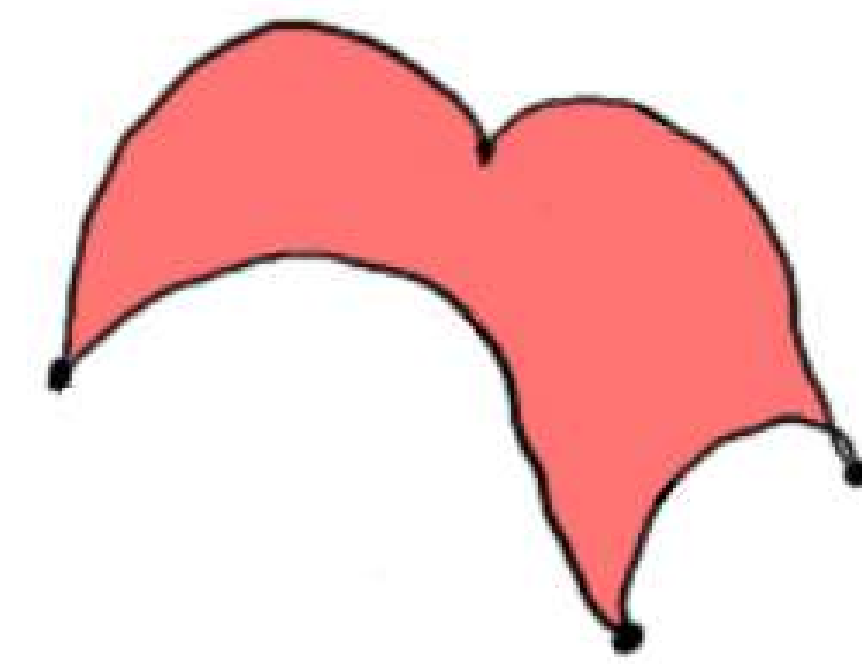
PARAMETRIC SKETCH

1.



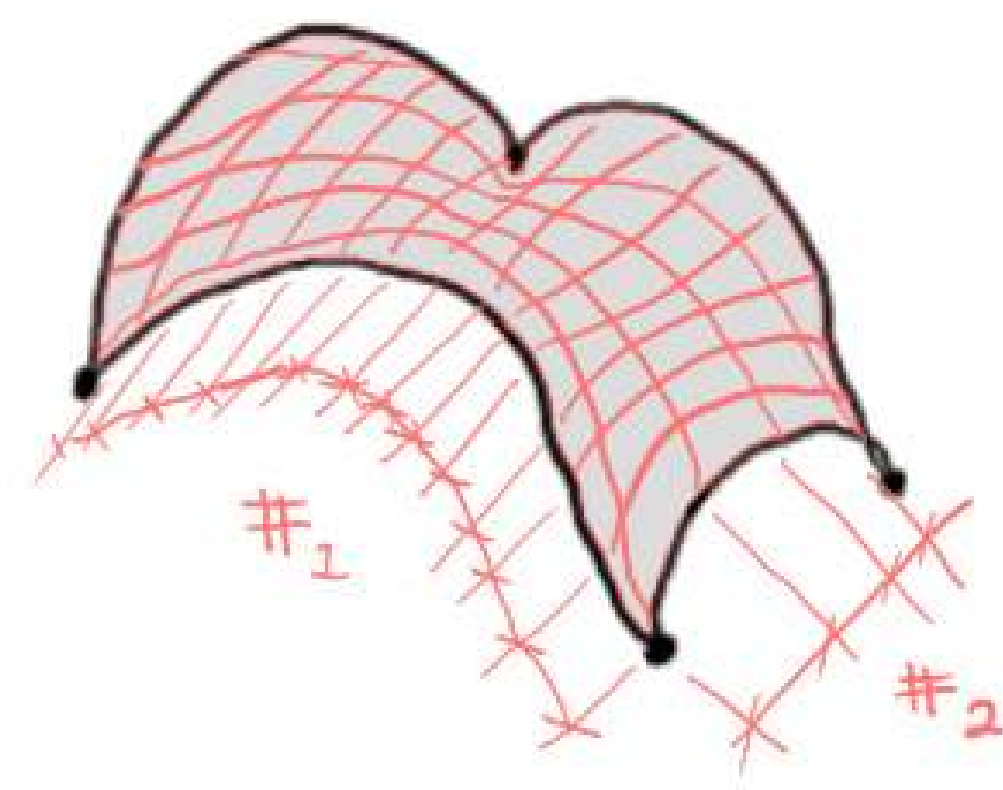
Set up four corner points.

2.



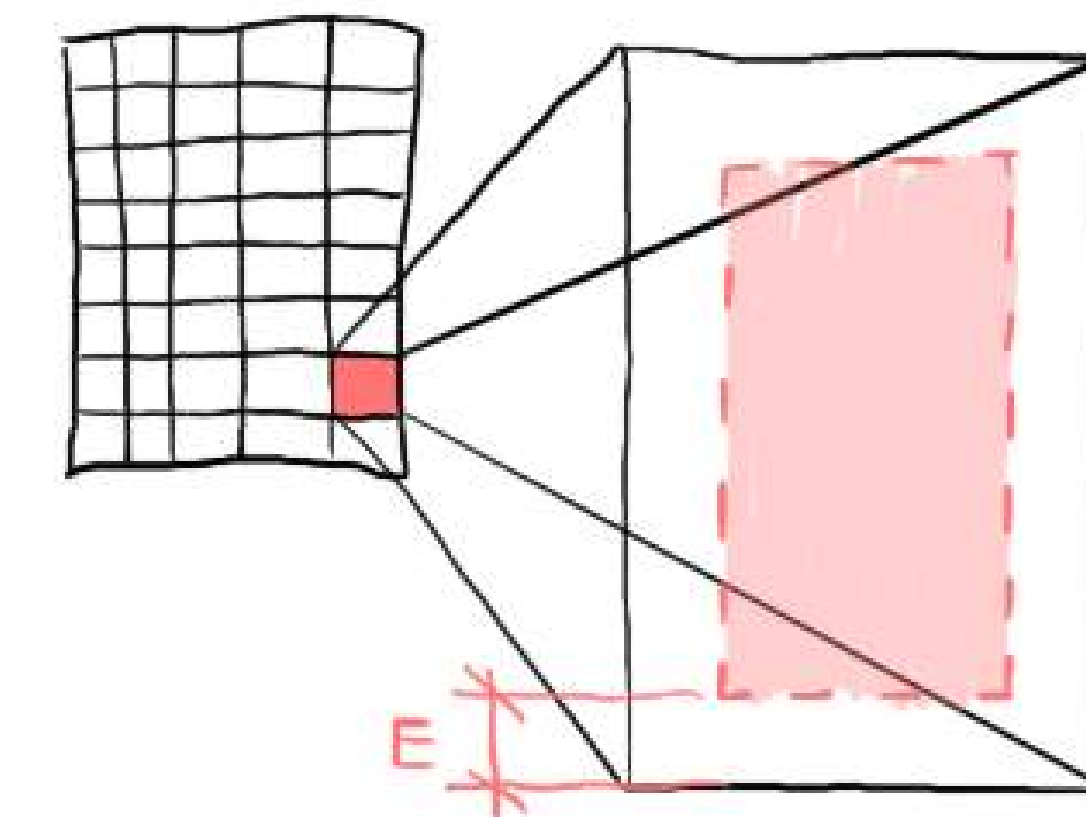
Create a surface from points.

3.



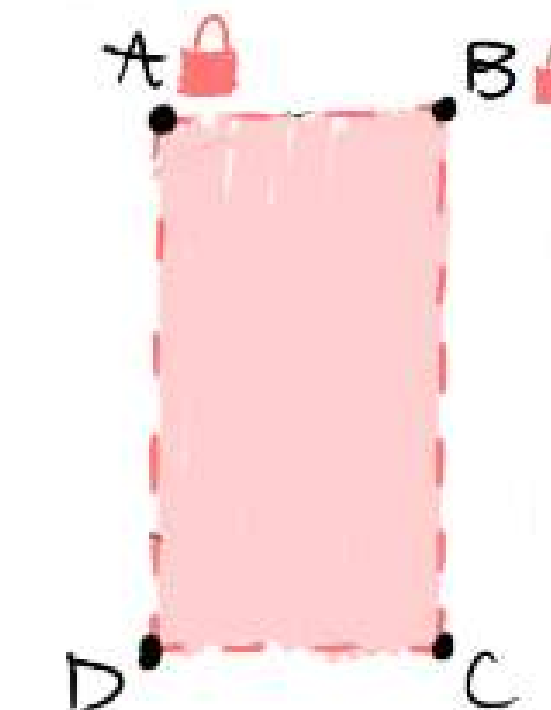
Divide surface by #1 & #2 values.

4.



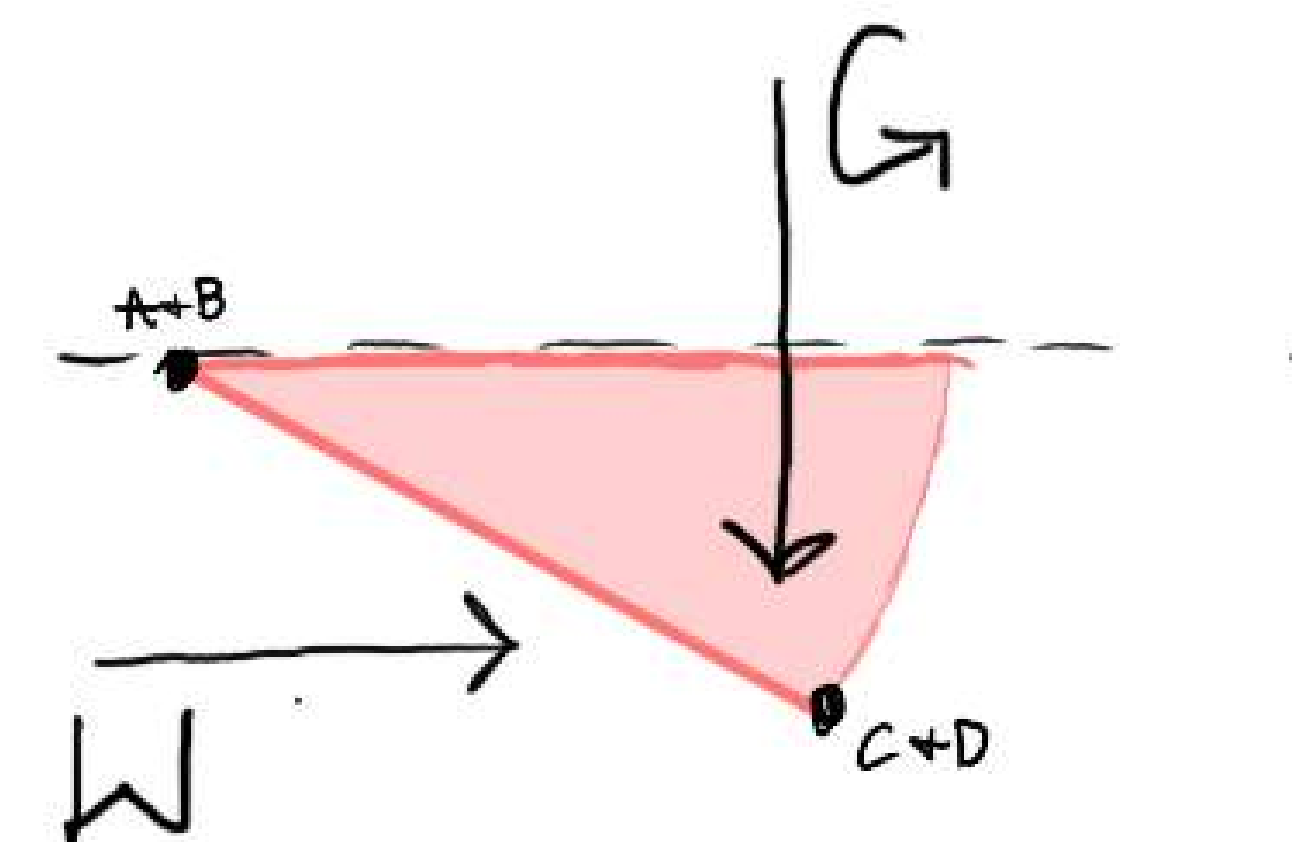
Isolate a panel and offset by value E.

5.



Fix corners A & B.

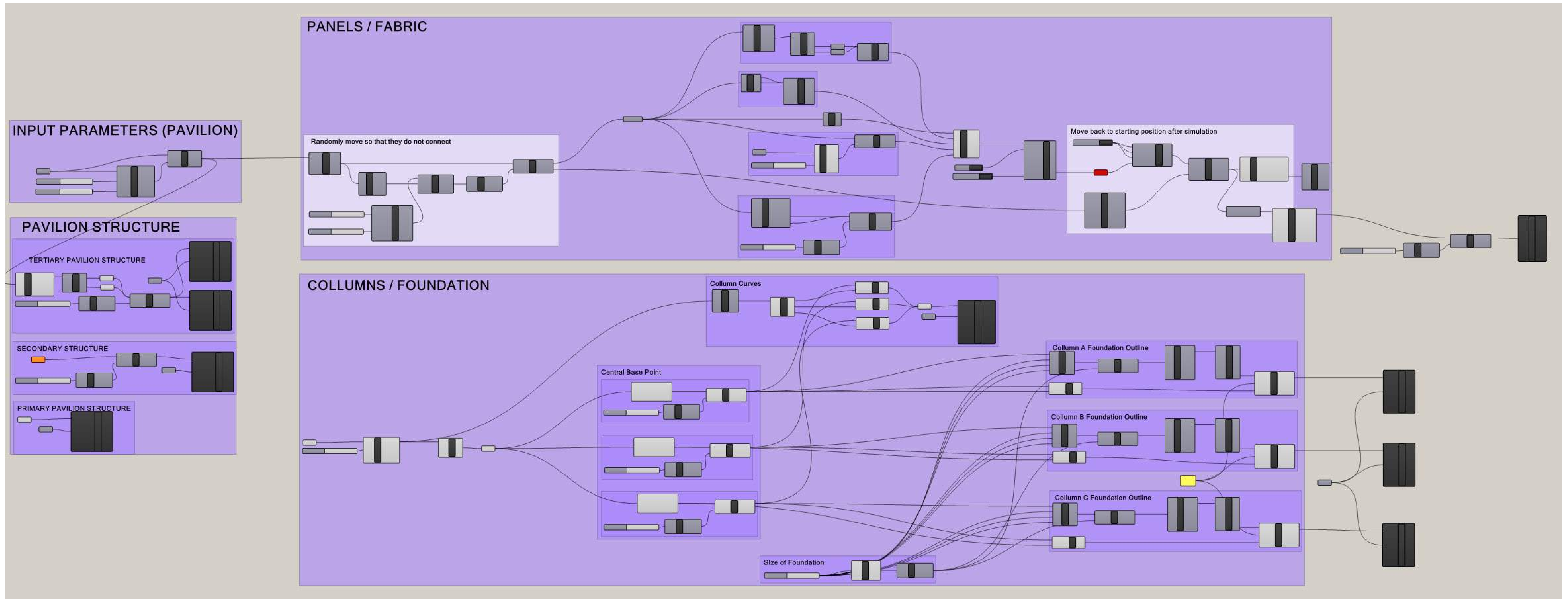
6.



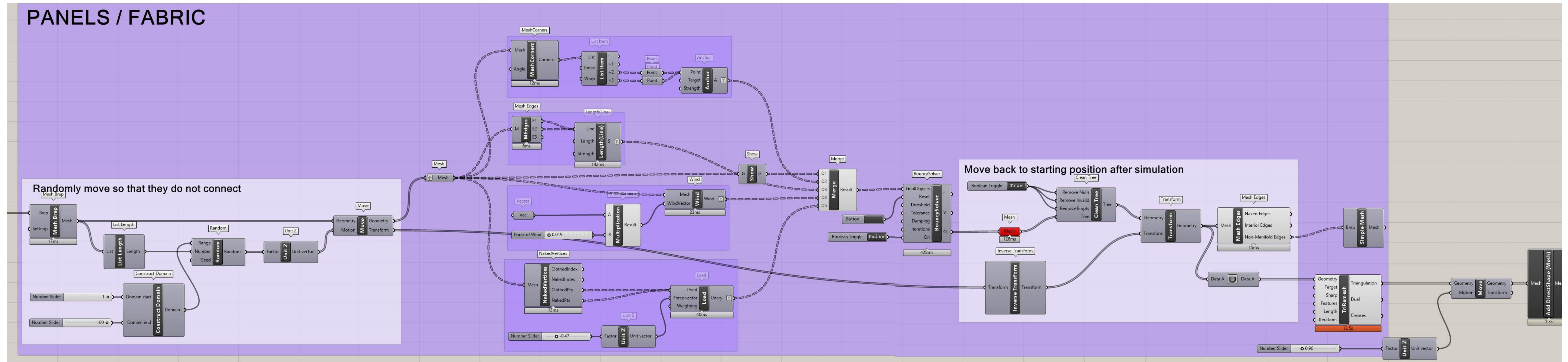
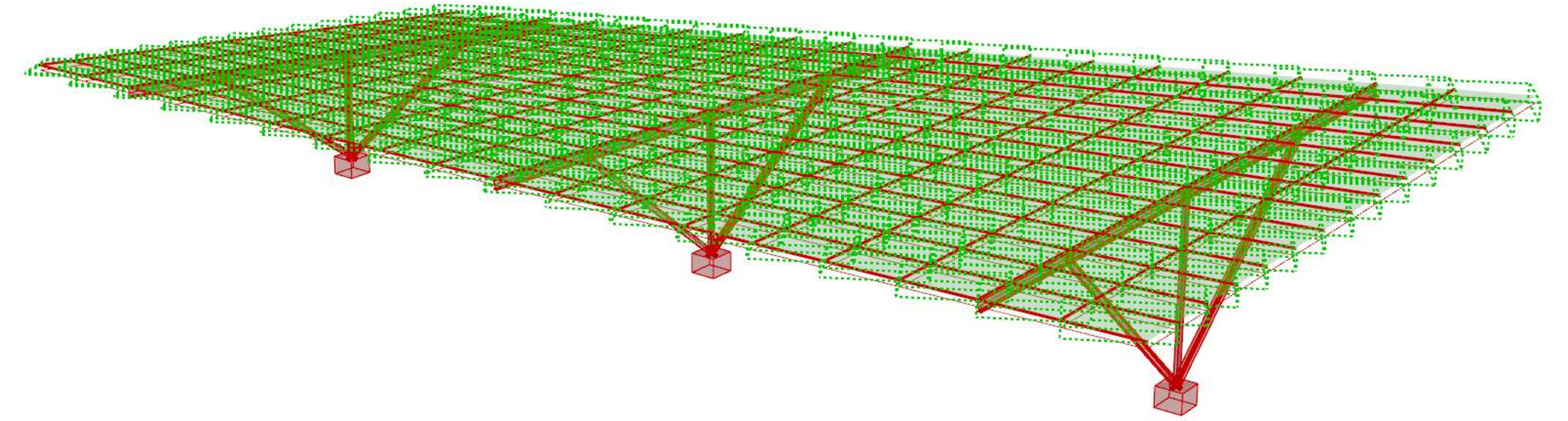
Add gravity (G) & Wind force (W) to panel.

MODELING

When modeling I broke the script into four main categories. Input, Panels, Beams, & Columns / Foundations.



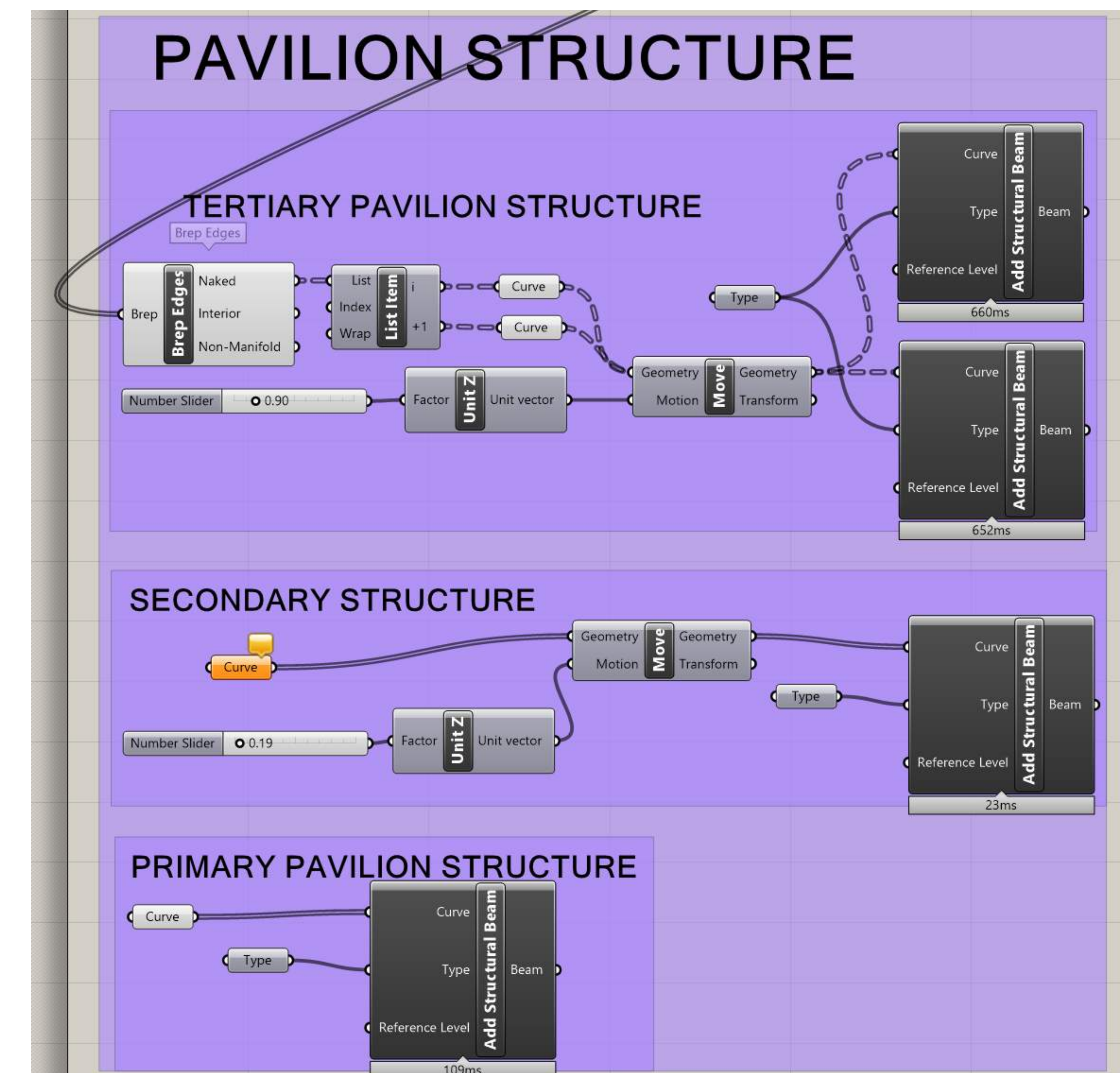
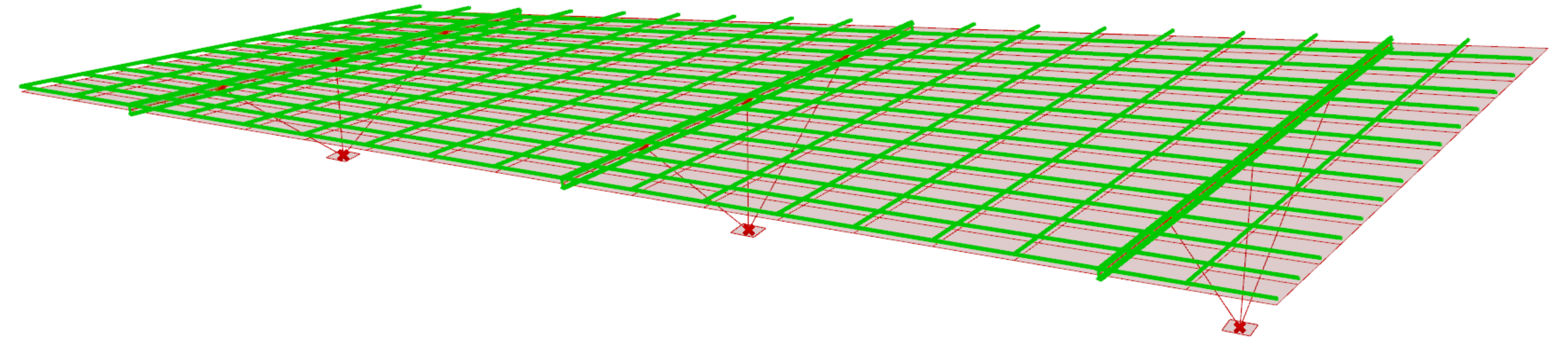
MODELING | PANELS / FABRIC



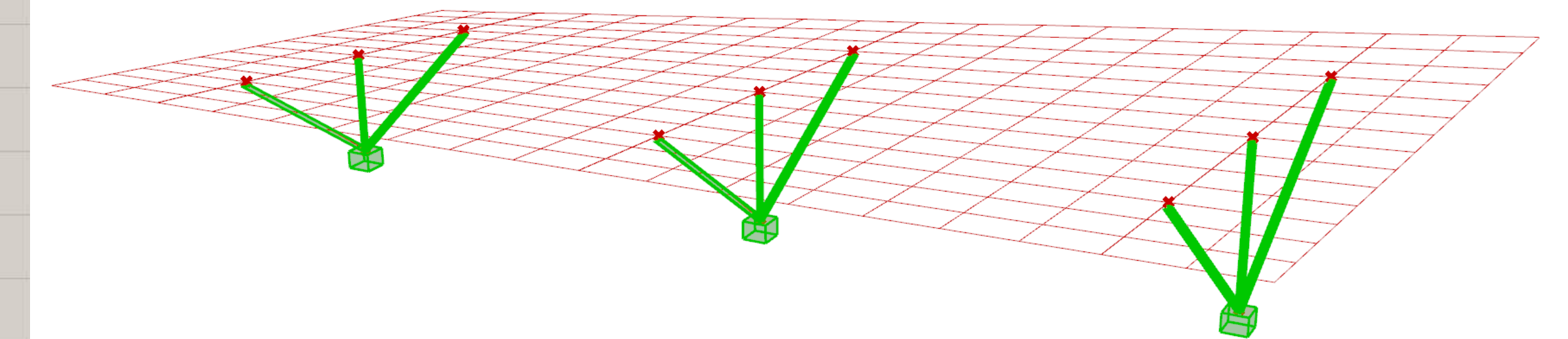
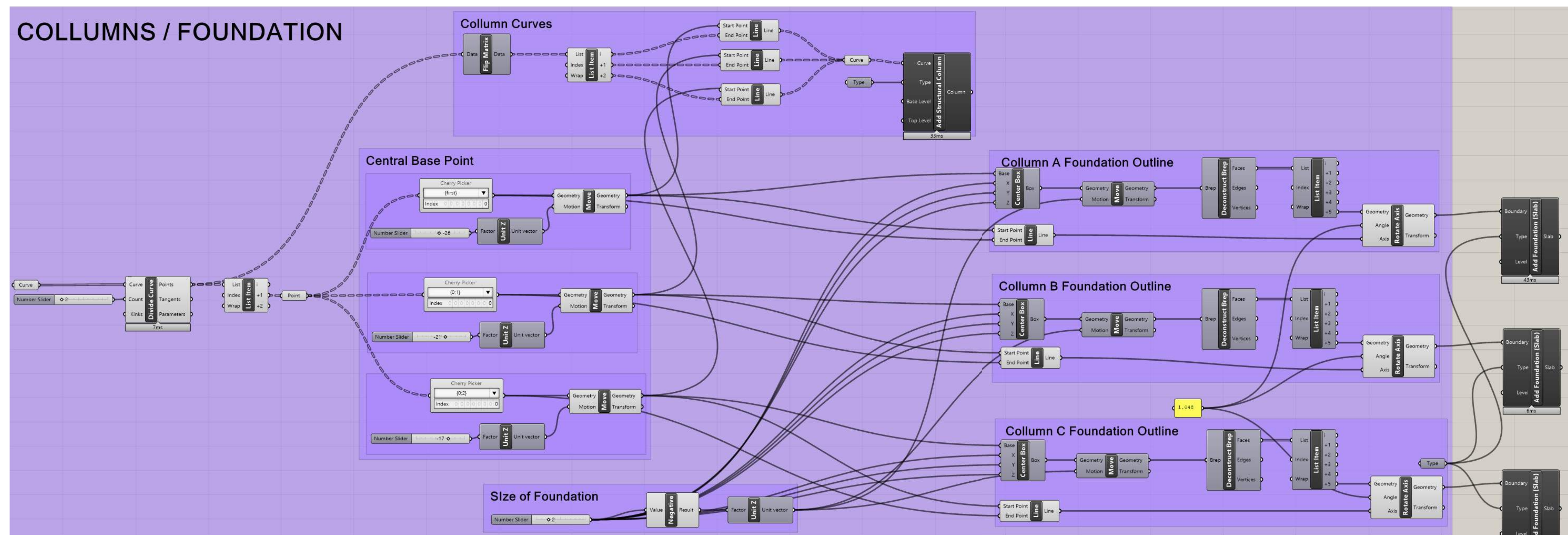
The panel / fabric script takes panels from a divided surface and runs them through a Kangaroo solver. Panels are anchored at two corners, boundaries are offset to avoid intersection, and wind and gravity forces (site specific) are applied. Randomize and move-back groups resolve a Kangaroo glitch so panels act independently. The final direct shape geometry is then brought into Revit for documentation.

MODELING | BEAMS

The beams are organized into three categories: primary, secondary, and tertiary. Primary members are the largest W-flange beams connected directly to the columns. Secondary beams, also W-flange, connect to the primary structure. The tertiary system consists of smaller metal frames that support the fabric panels.

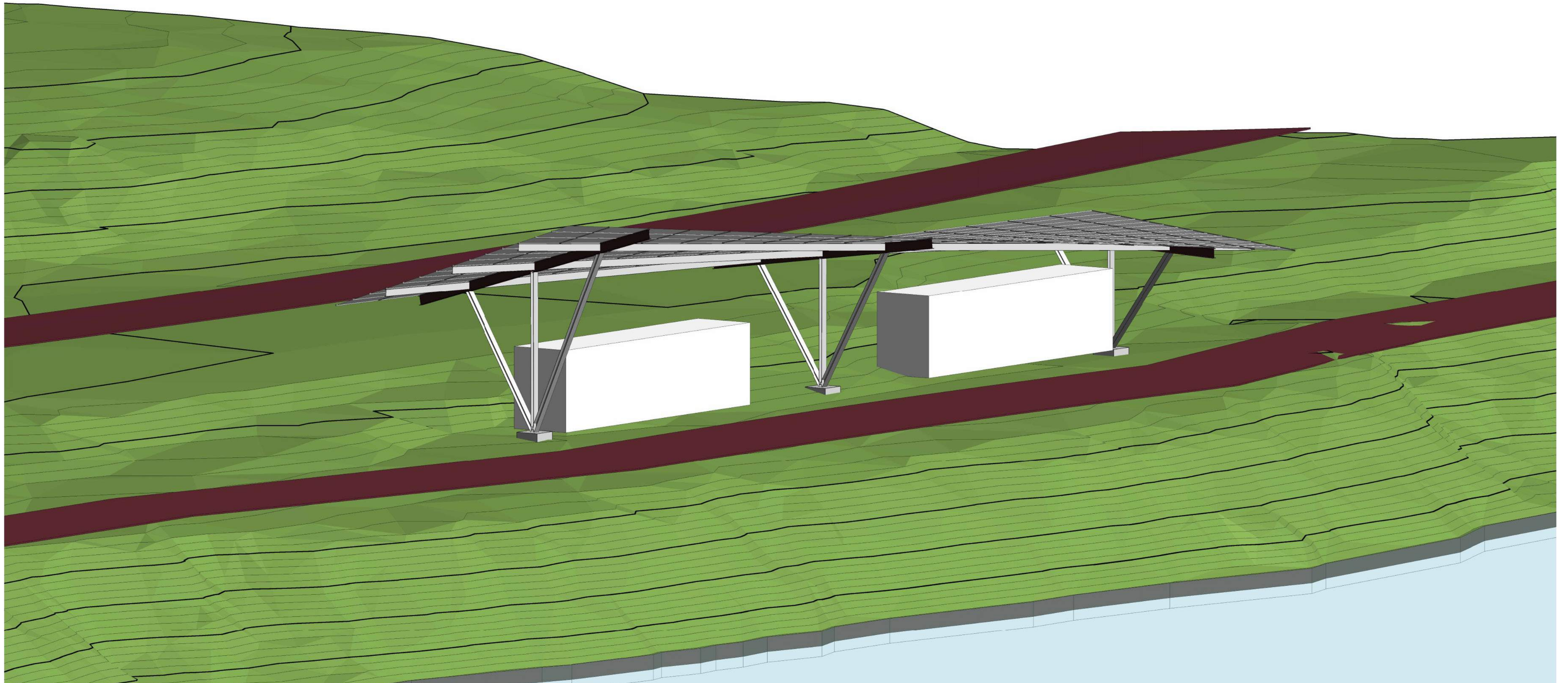


MODELING | COLUMNS & FOUNDATIONS



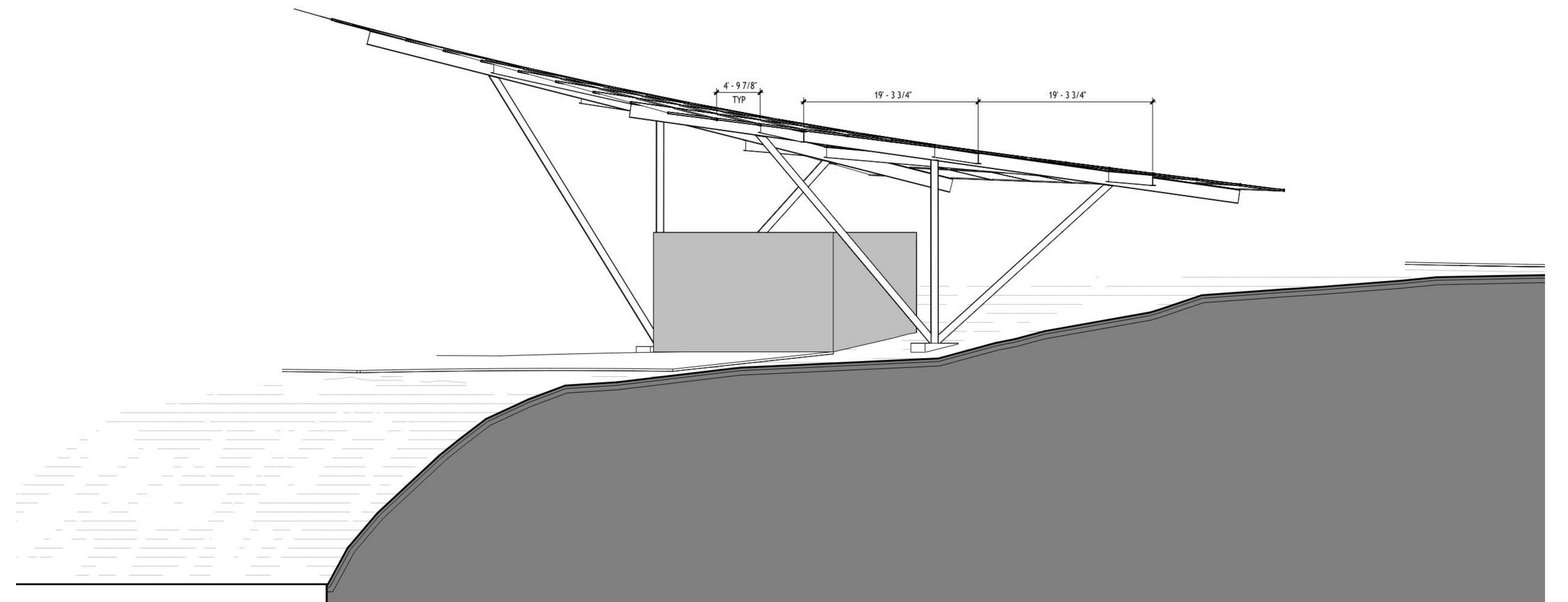
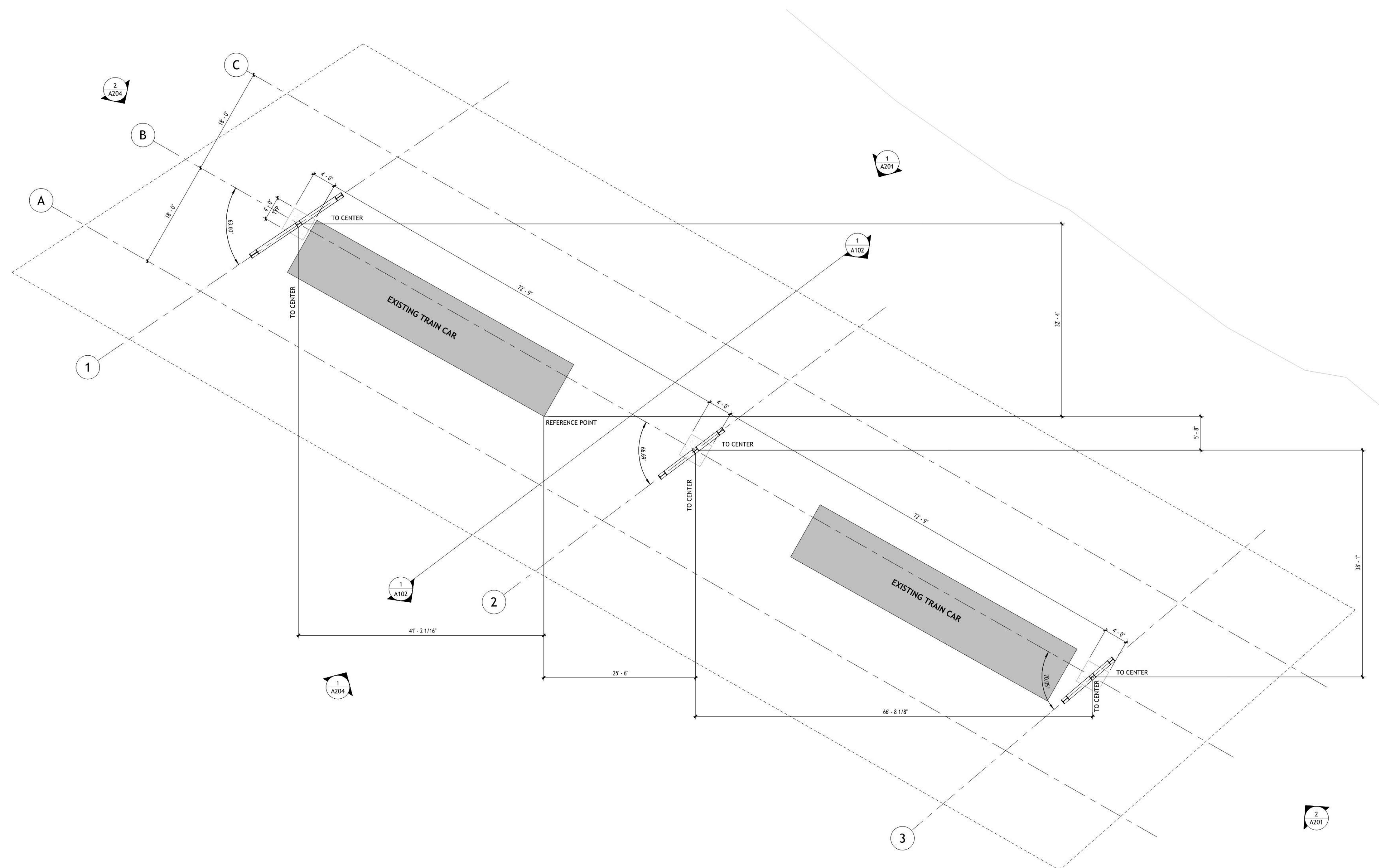
The final script divides beam curves into three points, selects the midpoint, and projects it to the ground to establish column lines. From this foundation center, a box is generated and its top outline isolated to define the slab perimeter.

DOCUMENTATION | 3D VIEW

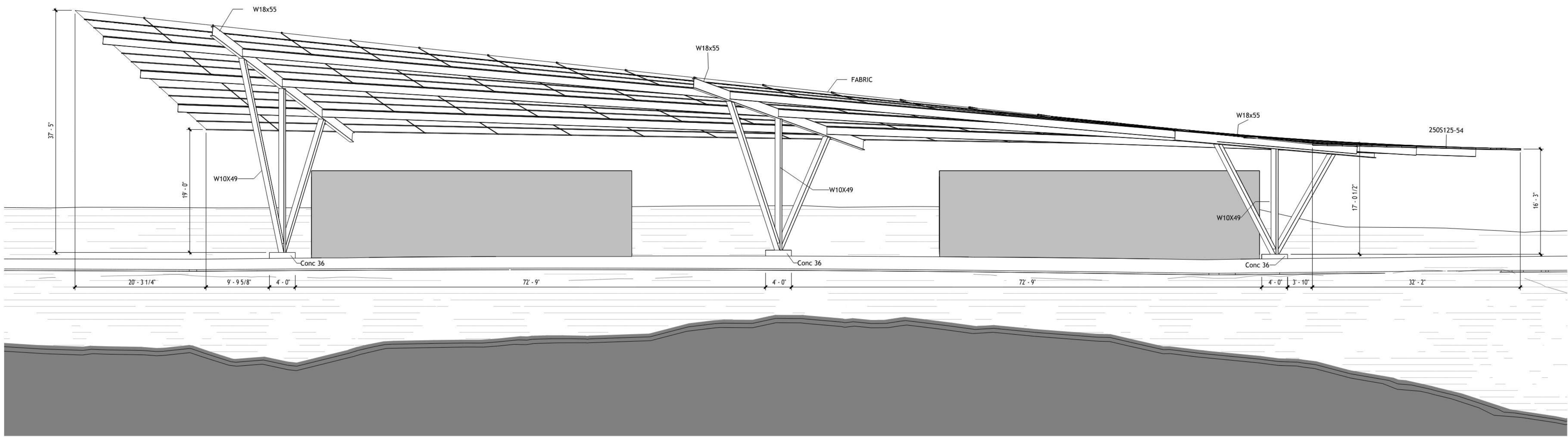


DOCUMENTATION | PLAN & SECTION

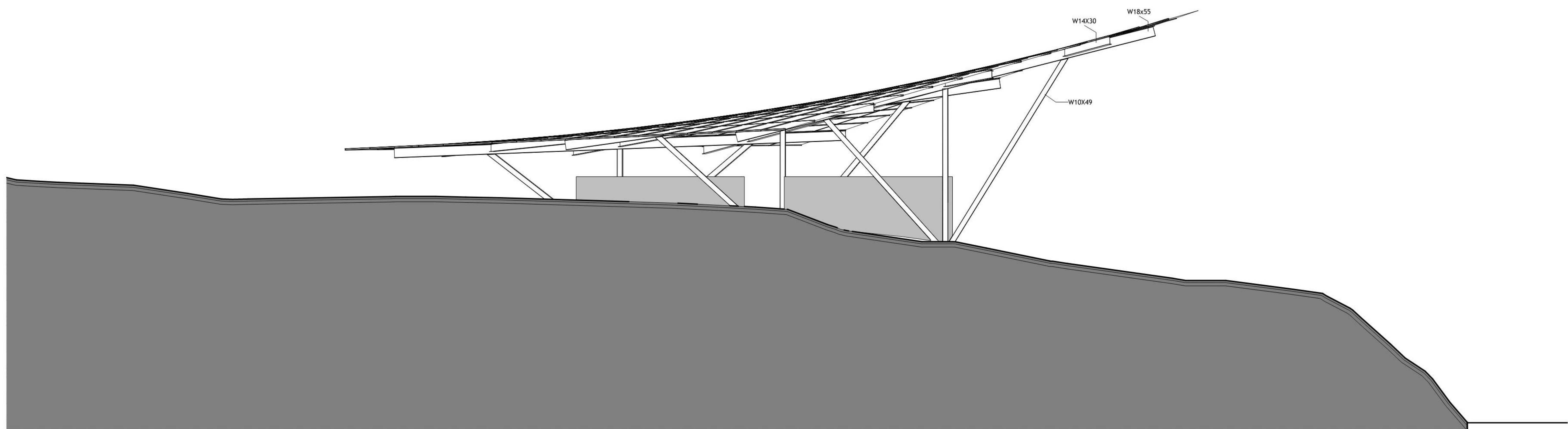
When I began documenting the design in revit, I quickly realized that while the current scheme effectively conveys my intent, it was not developed with documentation in mind. Many elements are not parallel, which complicates dimensioning and standard annotation. Thus in the plans and elevations, the dimensions deviate from typical conventions. This is something I would refine in a subsequent iteration of the design.



DOCUMENTATION | ELEVATIONS

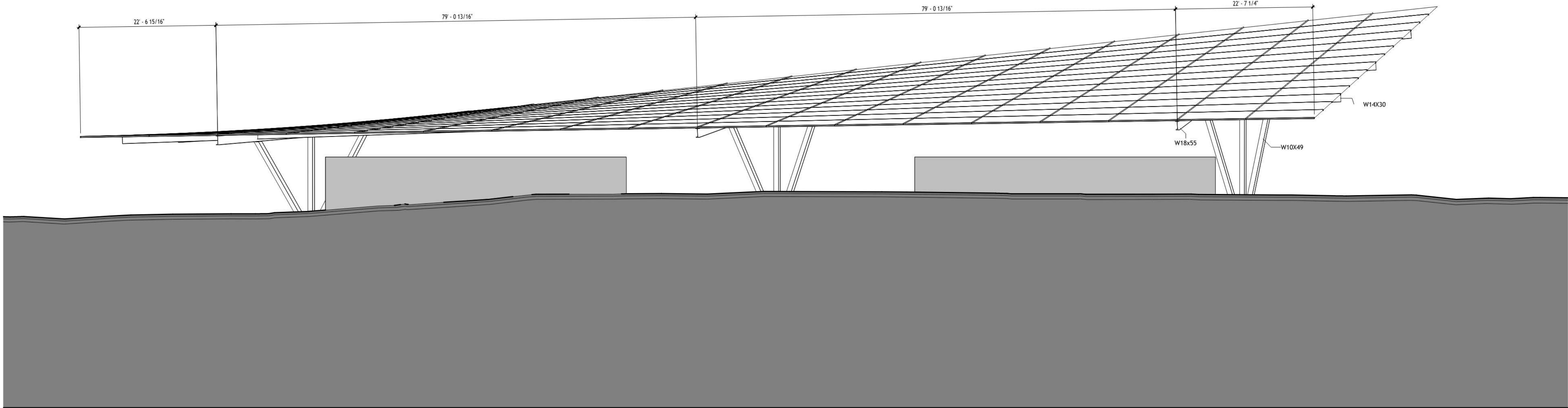


North Elevation

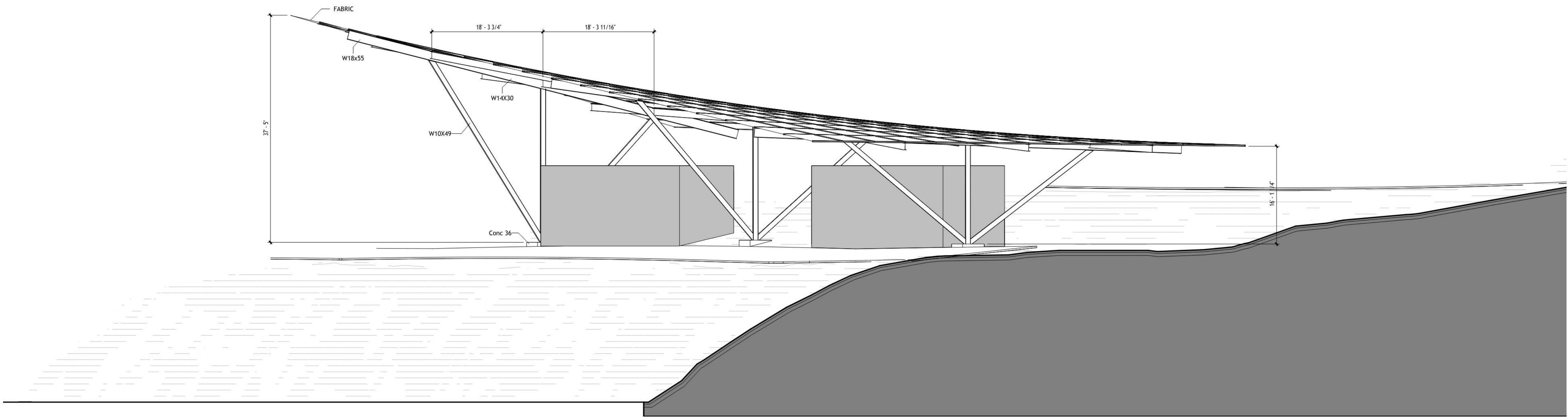


East Elevation

DOCUMENTATION | ELEVATIONS



South Elevation



West Elevation

RENDERING



Lighting Study: Sphere Test

A key factor in producing a strong rendering is accurate lighting. A simple way to evaluate this is the sphere test: placing chrome, black, gray, and white spheres in the scene reveals whether the image is too dark or overexposed.

BIRDS EYE RENDERING



INTERIOR RENDERING



EXTERIOR RENDERING

